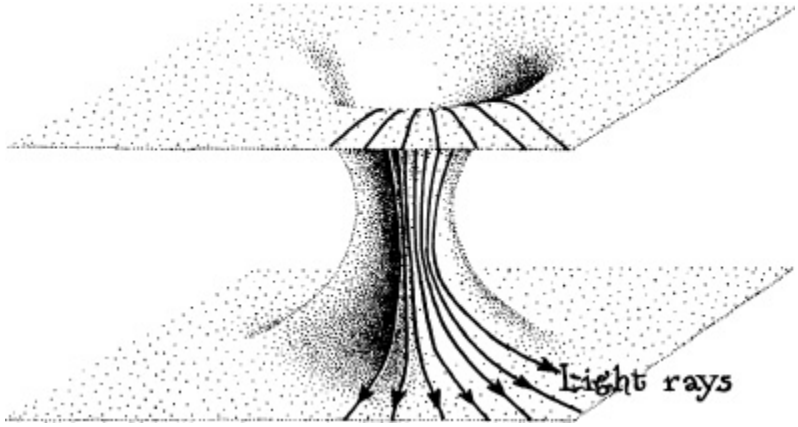
Box 14.1

Holding a Wormhole Open: Exotic Material

Any spherical wormhole through which a beam of light can travel will gravitationally defocus the light beam. To see that this is so, imagine (as drawn below) that the beam is sent through a converging lens before it enters the wormhole, thereby making all its rays converge radially toward the wormhole's center. Then the rays will always continue to travel radially (how else could they possibly move?), which means that when they emerge from the other mouth, they are diverging radially outward, away from the wormhole's center, as shown. The beam has been defocused.



The wormhole's spacetime curvature, which causes the defocusing, is produced by the "exotic" material that threads through the wormhole and holds the wormhole open. Since spacetime curvature is equivalent to gravity, it, in fact is the exotic material's gravity that defocuses the light beam. In other words, the exotic material gravitationally repels the beam's light rays, pushing them away from itself and hence away from each other, and thereby defocuses them.

This is precisely the opposite to what happens in a gravitational lens (Figure 8.2). There light from a distant star is focused by the gravitational pull of an intervening star or galaxy or black hole; here the light is defocused.

Perhaps our prejudice against the existence of exotic material is wrong, I thought to myself as I rode down Interstate 5. Perhaps exotic material *can*